

CS151 Data Structures

Quiz: 2
Name:

Date: 9/24/26

1. What is the difference between a linked list and an array? Explain in terms of memory organization as well as other significant operational differences.

2. Fill out the table with the appropriate big- O times. For both linked lists, assume that we maintain a head reference only (no tail).

	unsorted array	sorted array	unsorted linked list	sorted linked list
insert at beginning				
insert at end				
insert in interior (not beginning or end)				
delete at beginning				
delete at end				
delete in interior (not beginning or end)				
find min/max				
random access (read n th element)				
search				

3. Which operational times will change above if we also maintain a tail reference for the linked lists? List the times and what they will change to.

```

4. public class LinkedList<E> {
    private static class Node<E> {
        private E element;
        private Node<E> next = null;
        private Node<E> previous = null;
        public Node(E element) {this.element = element;}
        public Node<E> getNext() { return next; }
        public void setNext(Node<E> next) { this.next = next; }
        public Node<E> getPrevious() { return previous; }
        public void setPrevious(Node<E> previous) { this.previous = previous; }
        public E getElement() { return element; }
    }

    private Node<E> head;
    private int size = 0;

    public LinkedList() {
        head = null;
    }

    public void addFirst(E element) {
        Node<E> toAdd = new Node<E>(element);
        if (head != null) {
            head.setPrevious(toAdd);
        }
        toAdd.setNext(head);
        head = toAdd;
        size++;
    }
}

```

Given the starter code for a `LinkedList` class above, add a method `addMiddle` to `LinkedList` that inserts an element so that after insertion, the newly added node will be positioned in the middle of the list. If a list is even in length, we define middle as the first position of the second half of the list. Note also that although similar, this is a different linked list implementation from what you may have seen so far and you do not have access to any other methods except for what's listed above.